Reinforced Knowledge Distillation for Time Series Regression

Qing Xu, Keyu Wu, Min Wu, *Senior Member, IEEE*, Kezhi Mao, *Member, IEEE*, Xiaoli Li, *Senior Member, IEEE*, and Zhenghua Chen, *Senior Member, IEEE*

***Abstract*—As one of the most popular and effective methods in model compression, knowledge distillation (KD) attempts to transfer knowledge from single or multiple large-scale networks (i.e., *Teachers*) to a compact network (i.e., *Student*). For the multiteacher scenario, existing methods either assign equal or fixed weights for different teacher models during distillation, which can be inefficient as teachers might perform variously or even oppositely on different training samples. To address this issue, we propose a novel reinforced knowledge distillation method with negatively correlated teachers which are generated via negative correlation learning. The negatively correlated teachers would encourage teachers to learn different aspects of data and thus the ensemble of them can be more comprehensive and suitable for multiteacher KD. Subsequently, a reinforced KD algorithm is proposed to dynamically employ proper teachers for different training instances via dueling double deep Q-network (DDQN). Our proposed method complements the existing KD procedure on teacher generation and selection. Extensive experimental results on two real-world time series regression tasks clearly demonstrate that the proposed approach could achieve superior performance over state-of-the-art (SOTA) methods. The PyTorch implementation of our proposed approach is available at https://github.com/xuqing88/RL-KD-for-time-series-regression.**

***Impact Statement*—The advanced deep neural networks often suffer from complex network architectures which severely hinder their deployment on resource-limited devices. Knowledge distillation, as one of the most popular model compression techniques, aims to transfer the knowledge from single or multiple cumbersome teacher models to a compact student model. However, existing KD methods either laboriously select a particular teacher or simply assign equal or fixed weights for multiple teachers, resulting in tedious teacher selection procedure and poor distillation efficiency. Thus, in this article, we propose a novel reinforcement learning-based KD approach to overcome these limitations and our experimental results demonstrate that it consistently outperforms other SOTA methods on two real-world tasks. The proposed method can achieve average improvements of 5.7% and 15.7% in terms of RMSE and score, respectively, for machine RUL prediction task, and 8.1% in terms of mean localization error for indoor localization task, when comparing with SOTA methods.**

***Index Terms*—Knowledge distillation (KD), reinforcement learning (RL), time series regression.**

Manuscript received 10 April 2023; revised 31 August 2023; accepted 1 December 2023. Date of publication 18 December 2023; date of current version 21 June 2024. This work was supported by the National Research Foundation of Singapore, AME Young Individual Research Grant under Grant A2084c0167. This article was recommended for publication by Associate Editor Pierpaolo D'Urso upon evaluation of the reviewers' comments. *(Corresponding author: Zhenghua Chen.)*

Qing Xu and Xiaoli Li are with the Institute for Infocomm Research, A*STAR, Singapore and with the School of Electrical and Electronic Engineering, Nanyang Technological University, 138632, Singapore (e-mail: Xu_Qing@i2r.a-star.edu.sg; xlli@i2r.a-star.edu.sg)

Keyu Wu, Min Wu, and Zhenghua Chen are with the Institute for Infocomm Research, A*STAR, 138632, Singapore (e-mail: Wu_Keyu@i2r.a-star.edu.sg; wumin@i2r.a-star.edu.sg; chen0832@e.ntu.edu.sg).

Kezhi Mao is with the School of Electrical and Electronic Engineering, Nanyang Technological University, 639798 Singapore (e-mail: EKZMao@ntu.edu.sg).

Digital Object Identifier 10.1109/TAI.2023.3341854

I. Introduction

TIME series regression has many potential use-cases in Internet of Things (IoT), such as indoor localization [1], occupancy estimation [2], and remaining useful life (RUL) prediction [3], [4]. Recently, deep neural networks (DNNs) have exhibited remarkable capabilities in time series regression tasks. However, those DNNs with state-of-the-art (SOTA) performance often consist of billions of model parameters, which cannot be fully utilized considering that time series regression algorithms generally require to be deployed on IoT devices with limited computational resources. To address this issue, various cutting-edge deep model compression techniques have been developed to improve model efficiency without significantly compromising model accuracy, e.g., network pruning and quantization [5], network architecture search (NAS) [6], and knowledge distillation (KD) [7], [8]. Among them, KD related approaches have drawn enormous attentions due to their effectiveness and flexibility on transferring knowledge from a complex teacher model to a compact student model.

In most of existing KD approaches, the knowledge either comes from a single teacher [9] or comes from an ensemble of multiple teachers [7]. For single teacher distillation scenario, it is common to employ a pretrained large scale model with SOTA performance as the teacher. However, numerous empirical experiments have shown that a stronger teacher does not necessarily stand for a better teacher [10] due to the significant capacity gap between it and the compact student. Therefore, selecting a proper teacher could be very time-consuming and tedious in many applications.

Alternatively, one can use the ensemble of multiple teachers to omit the teacher selection procedure. On the one hand, it is common to train several models independently [11] or sequentially [12] to get multiple teachers and then combine them together as the ensemble. One limitation of these two methods is that there is no feedback from the combination stage to the individual training stage, which limits their performance. Moreover, the success of distilling knowledge for cross-modal

2691-4581 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

data [13], [14] and multiview data [15], [16] in recent works has shown the necessity of diversifying teachers in the ensemble as it would provide different views of data for the student. On the other hand, simply taking average [7] or using fixed weights [15], [17] over all models in the combination stage could also be inefficient. The teacher models would generally converge to different local minima due to random initialization, different optimization trajectories, or other factors. It is possible that some teachers would contribute less or even have the negative contributions to the performance on certain instances [18]. Hence, in order to achieve better distillation performance, it is necessary to dynamically select proper teachers for different instances during the entire distillation process.

Inspired by the above insights, we systematically study how to generate diverse teachers via designing a negative correlation learning (NCL) strategy and proposing a reinforced knowledge distillation approach to dynamically select different teachers during distillation via reinforcement learning (RL). We verify the effectiveness of our approach on time series regression tasks which are commonly needed in flexible model compression techniques. In particular, we follow the same problem setup as [19], [20] and target at effectively transferring knowledge between disparate network architectures (e.g., from LSTM-based teachers to a CNN-based student). We expect the CNN-based student to be capable of capturing the temporal information buried in the sequential data like LSTM but with less model complexity like CNN, thereby more friendly for deployment.

Our contributions are summarized as follows.

1) We propose a reinforced KD approach based on reinforcement learning to dynamically select proper teachers for different instances, which significantly enhances the efficacy of multiteacher KD.
2) We propose to generate highly diverse teachers for efficient learning in multiteacher KD via a NCL module.
3) Extensive experiments on two time series regression tasks have demonstrated that the proposed method significantly outperforms other SOTA approaches.

The rest of the article is organized as follows: Section II reviews some related works on multiteacher KD approaches and several works combining reinforcement learning with knowledge distillation. Section III introduces the details of our proposed method. In Section IV, the experimental setup and results are presented, followed by ablation study and sensitivity analysis. Section V concludes the article and demonstrates some potential future works.

II. Related Work

Hinton et al. were the first to term the soften logits of a cumbersome model as knowledge and then distilled it to a compact model [7]. Romero et al. further extended the knowledge from soften logits to the output of intermediate layers (i.e., feature distillation) in [9]. After that, various KD methods, such as [19], [20], [21], [22], [23], have been proposed to distill knowledge from a single teacher in various applications. However, the selection of teacher model is not explicitly stated in these works and it is commonly based on personal prior experience.

Considering the fact that better performance is often achieved by the ensemble of DNNs, several multiteacher KD approaches have recently been proposed. Intuitively, taking average of outputs of all teachers is a good choice and has proved to be effective in [7], [13], [24]. Besides, Chebotar and Water adopted fixed weights for different teachers and used the grid search to identify the optimal weights [17]. Wu et al. also employed fixed weights for a three-teacher distillation on a video action recognition task but did not clearly state how to determine these weights [15]. Nonetheless, simply combining all teaches with equal or fixed weights is insufficient as each teacher may perform very differently from other teachers on certain samples. Fukuda et al. proposed two strategies for multiteacher KD scenario. One is called switched-training which randomly selects a model from teachers' pool to guide student's training, the other one is called augmented-training which respectively updates student's parameters with the outputs of each teacher [25]. Our experimental results show that the switched-training method even underperforms the averaging ensemble and the augmented-training could be regarded as an extreme case of our proposed methods. Our proposed method is able to adaptively select proper teachers for different training instances and thus significantly boost student's performance via distillation.

Another research hot-spot is to combine RL with knowledge distillation. Fan et al. proposed a policy gradient-based RL method to assign different weights to samples within homogeneous classes during distillation to handle the problems of multiclass imbalanced classification [26]. Wang et al. utilized the RL strategy to fine-tune the structure of a generative model, which is first simplified via the KD, to enhance the diversity of generated molecules [27]. Tsantekidis et al. trained the RL-based financial trading teacher agents under different environments and then distilled the knowledge to a student to improve its training reliability [28]. Liang et al. presented a RL-based framework to select appropriate unlabeled samples from target language when transferring knowledge from the source language [29]. Our work differs from all the aforementioned works in terms of how to integrate RL techniques and KD approaches.

One highly related work to ours is [30], which tried to assign different weights to multiple teacher models during knowledge transferring to the student. However, our work is different from it in the following aspects. First, our problem formulation is totally different, not only in terms of the definition of state and reward, but also in the reinforced learning strategy. Specifically, Yuan et al. [30] adopted the standard policy gradient method while we propose an off-policy DQN algorithm to optimize the RL model. Our method is able to achieve more sample efficiency through reusing the collected experiences. Moreover, the Monte-Carlo based policy gradient method adopted in [30] uses an estimation of gradient generated by only a series of data during each gradient update. Therefore, their noise estimation can adversely affect the learning stability, while our DQN-based method can lead to more stable performance via target network updates. Second, to the best of our knowledge, we are the first to exploit negative correlation learning to generate highly diverse teachers for the multiteacher KD scenario. With the proposed RL-based teacher selection method, the negatively correlated

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

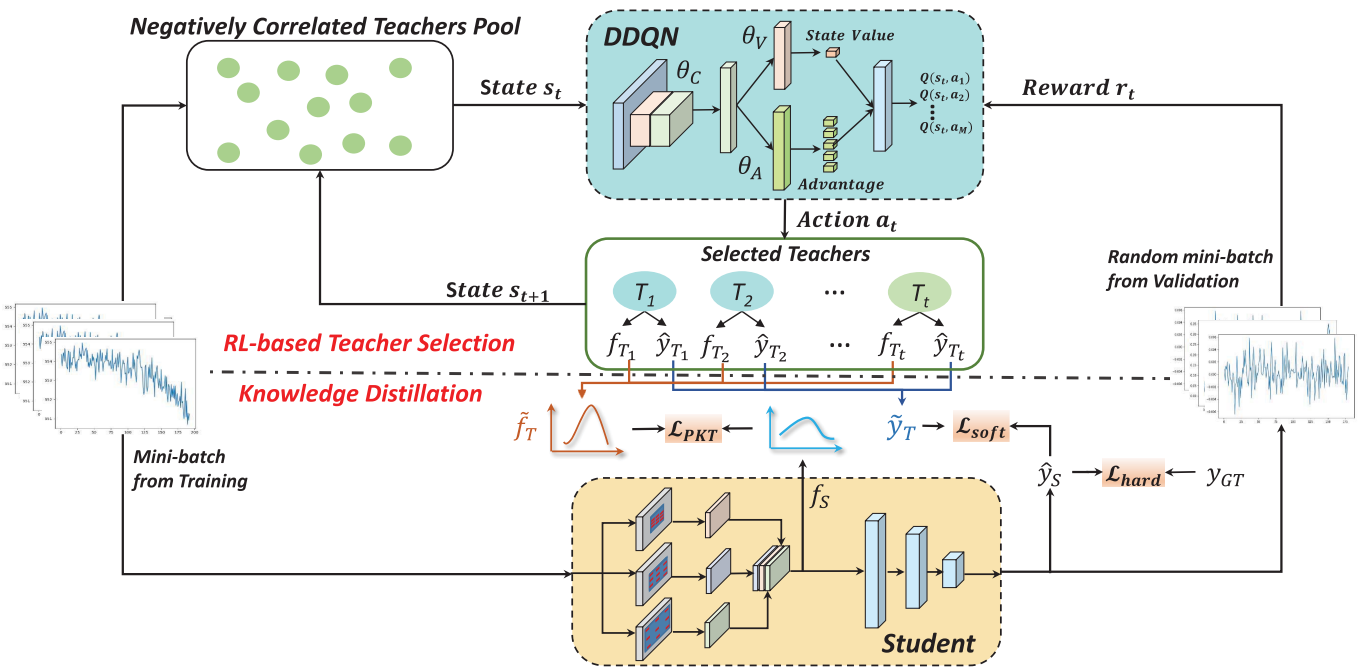

Fig. 1. Proposed RL-based KD framework. The upper part illustrates the RL-based teacher selector. We sample out one teacher from the pool at each step and aggregate them for knowledge transfer. The lower part is the KD process. With the selected teachers, the student network is optimized via minimizing the loss which is the combination of probabilistic knowledge transfer (PKT) loss, soft loss, and hard loss.

teachers are more efficient in student's learning than other types of teachers, such as independent teachers used in [30].

III. Methodology

In this section, we first introduce the negatively correlated teachers generated by designing a NCL module. Then, a RL-based teacher selection method is presented. Lastly, the distillation is performed between the selected teachers and the student. The overall framework for our proposed approach is illustrated in Fig. 1.

A. Negatively Correlated Teachers

As aforementioned, for independent- and sequential-trained models, there is no interaction between teacher models during training stage, as well as no feedback from the combination stage. Contrarily, negative correlation learning strategy [18] emphasizes the interaction and cooperation among individual networks and encourages each network to learn different aspects of training data by introducing a penalty term into individual network training process. Specifically, for the ith network in the ensemble, the loss function is defined as (1). The first term is the empirical training loss function, e.g., *mean square error*, between the ground truth y and the prediction $\hat{y}_i$ from the ith teacher. The second term is the correlation penalty and λ is a hyperparameter to adjust the strength of the correlation penalty. Here, $\tilde{y}$ is the average of predictions of all models in the ensemble

$$\mathcal{L}_i = (\hat{y}_i - y)^2 + \lambda * (\hat{y}_i - \tilde{y}) * \sum_{j \neq i} (\hat{y}_j - \tilde{y}). \tag{1}$$

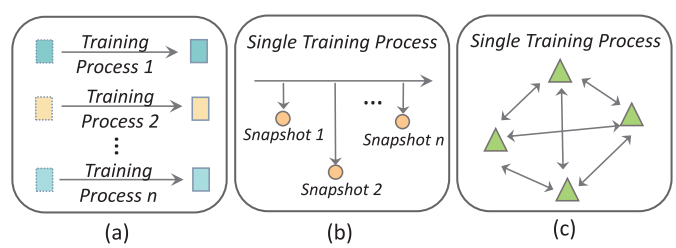

Fig. 2. Different methods to get teacher models: (a) each teacher is independently trained; (b) teachers are sequential snapshots over single training process; and (c) teachers are simultaneously trained via negative correlation learning.

By minimizing the penalty term, the individual's error is negatively correlated to the errors of the rest in the ensemble, so that the diversity of the ensemble would be increased. Our experimental results show that comparing with teachers via independently or sequentially training, negatively correlated teachers are more knowledgeable for transferring the comprehensive knowledge to a student. Fig. 2 illustrates the above mentioned three different methods to generate multiple teachers.

B. RL-Based Teacher Selection

When multiple teachers are available in teacher–student learning paradigm, since each individual teacher model may have inconsistent performance on different samples or mini-batches, dynamically employing proper teachers on instance level or mini-batch level could provide better understandings

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 RL-based Teacher Selection

Input: Negatively correlated teachers pool with M models. Student with Θ_s. An online network $\mathcal{Q}$ and a target network $\mathcal{Q}'$ initialized with Θ_q and Θ_q^-, respectively; Training data $\mathcal{D}$ and validation data $\mathcal{V}$; Epoch number L; Step number K; Replay Buffer $\mathcal{H}$.

for epoch $l \in [1, L]$ **do**
  Randomly shuffle $\mathcal{D}$
  for each mini-batch $\mathcal{D}_b \in \mathcal{D}$ **do**
    Calculate the initial state for $\mathcal{D}_b$
    for step $t \in [1, K]$ **do**
      Get state s_t and sample action $a_t \sim \mathcal{Q}(s_t)$.
      Pick out a teacher from the pool based on a_t.
      Update next state s_{t+1}.
      Update Θ_s with selected teachers via KD.
      Calculate r_t using a random batch from $\mathcal{V}$.
      Set $\beta = 0$ if step end, otherwise $\beta = 1$
      Store the tuple (s_t, a_t, r_t, s_{t+1}, β) to $\mathcal{H}$.
    end for
  end for
  Update $\mathcal{Q}$ and $\mathcal{Q}'$ via Algotithm 2.
end for

for specific data and thus effectively boost student's performance. Fig. 1 illustrates the overview of the proposed RL-based teacher selection process. Particularly, we utilize the dueling double deep Q-network (DDQN) [31] to obtain the optimal teacher selection policy as it can effectively mitigate potential overestimation problem via dueling architectures and address instability issue via techniques like target network and memory replay. For every training batch, we sample out one teacher from the negatively correlated teachers pool at each step. Then, the selected teacher at the current step participates in the KD stage together with the other teachers sampled out at previous steps. The averaged outputs of intermediate layer and last layer of the selected teachers are regarded as the knowledge to guide student's training. Algorithm 1 presents the details of the proposed RL-based teacher selection process. In the following, the definitions of the state, action, reward and the optimization of DDQN are introduced in details.

State. In our proposed method, the state $s_t \in \mathbb{R}^{n_b \times k \times M}$ at any step $t \in [1, K]$ is formulated as the aggregation of feature maps from the intermediate layer of all negatively correlated teachers. Here, n_b is the number of samples in a mini-batch $\mathcal{D}_b \in \mathcal{D}$, k is the dimension of flattened feature maps from LSTM-based feature extractor, and M is the total number of negatively correlated teachers in the pool. Note that after a teacher is sampled out from the pool based on the optimal policy, the next state s_{t+1} is generated by zeroing out the corresponding feature maps in s_t. In other words, we mask off those teachers who have already been selected out in previous steps by forcing the values of their flattened feature maps in state s_t to be zero. To save computational cost, we only perform one-time inference for all negatively correlated teachers in mini-batch $\mathcal{D}_b$ as the initial state and conduct the zeroing-out operation at each step.

Action. For a specific teacher i in the pool, there are only two actions $a \in \{0, 1\}$ associated with it, selecting or not for current mini-batch. Specifically, $a_t^i = 1$ means to select teacher i at current step t and $a_t^i = 0$ means not to select it. Moreover, since we only select one teacher at each step and the output of the DDQN is a set of Q-values, the optimal action a^* at state s_t is then determined by the following equation:

$$a^* = \underset{a}{\operatorname{argmax}}\, Q(s_t, a; \Theta_q). \tag{2}$$

Note that the argmax operation will only be applied to the remaining teachers in the pool.

Reward. The reward plays a crucial role on the learning of teacher selection policy as it could provide important feedback to DDQN for the value of choosing a particular action in current state. Motivated by [30], we use the student's performance on validation set to calculate the reward. However, we measure the linearized relative improvement of student's performance before and after distillation on a random batch from validation set every step as follows:

$$r = \max\Big(-1, \min\Big(\tanh\Big(4 * \frac{P' - P}{P'}\Big), 1\Big)\Big). \tag{3}$$

Here, P' and P represent the student's performance, e.g., root mean square error (RMSE), before and after optimization with KD at one training step, respectively. We adopt the tanh function to linearize the relative improvement on a random validation batch. Note that a positive value of tanh means the performance get improved (e.g., smaller RMSE). On the other hand, it is possible that the numerical value of relative improvement on RMSE is very small on a single optimization step due to a small learning rate. Therefore, we multiply the relative improvement by a coefficient to assign a good reward even when the improvement is marginal. What's more, we bound the reward between -1 and 1 so that the DDQN would be able to better converge.

Optimization of DDQN. The double DQN consists of two deep Q-networks, i.e., an online network $\mathcal{Q}$ with parameters Θ_q and a target network $\mathcal{Q}'$ with parameters Θ_q^-. $\mathcal{Q}$ and $\mathcal{Q}'$ share identical architecture and the initial Θ_q^- is a copy of Θ_q. As shown in Fig. 1, the DDQN network consists of three subnetwork: a common CNN encoder with parameters θ_C which takes a 3-D state matrix as input and outputs a vector, two separate streams (i.e., two stacked fully connected layers parameterized with θ_V and θ_A, respectively) to estimate the state-value $V(s; \theta_C, \theta_V)$ and advantages $A(s, a; \theta_C, \theta_A)$ for each action. Finally, the two streams are aggregated by the following equation to produce Q-values:

$$Q(s, a; \theta_C, \theta_V, \theta_A) = V(s; \theta_C, \theta_V) + A(s, a; \theta_C, \theta_A) - \frac{1}{2}\sum_{a'} A(s, a'; \theta_C, \theta_A). \tag{4}$$

Algorithm 2 presents the details of the optimization process for the DDQN. To be specific, a random batch of entries (s_t, a_t, r_t, s_{t+1}, β) is sampled out from replay buffer $\mathcal{H}$. Then, Θ_q is

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2 Optimization of DDQN

Input: An online network $\mathcal{Q}$ and a target network $\mathcal{Q}'$ initialized with Θ_q and Θ_q^-; Replay Buffer $\mathcal{H}$.

1: Randomly sample a mini-batch of $(s_t, a_t, r_t, s_{t+1}, \beta)$ from $\mathcal{H}$.
2: Calculate the estimated Q-value $Q(s_t, a_t; \Theta_q)$ via (4).
3: Calculate the target Q-value via (5).
4: Calculate the Huber loss between the estimated Q-value and target Q-value.
5: Update Θ_q via minimizing the Huber loss.
6: Update Θ_q^- via (6).

optimized via minimizing the Huber loss between the estimated Q-values and the target Q-values which are calculated from the following equation:

$$Q_{\text{tar}} = r_t + \beta * \text{done} * Q\Big(s_{t+1}, \underset{a_{t+1}}{\text{argmax}}\, Q(s_{t+1}, a_{t+1}; \Theta_q); \Theta_q^-\Big). \tag{5}$$

Here, $\beta \in \{0, 1\}$, $\beta = 0$ means next step $t+1$ is the terminal step and 1 means the opposite. $\gamma \in [0, 1]$ is the discount factor that provides a tradeoff between the immediate and future rewards. Lastly, the parameters Θ_q^- of target network $\mathcal{Q}'$ is updated via a moving average as shown in the following equation:

$$\Theta_q^- \leftarrow \delta * \Theta_q^- + (1 - \delta) * \Theta_q. \tag{6}$$

C. Knowledge Distillation

LSTM-based networks have demonstrated superior performance over CNN-based networks on time series data analytics [32]. Yet, they are generally more complex than CNN-based networks. In this work, we intend to distill knowledge from a powerful and complex LSTM to an efficient CNN for time series regression. Here, the knowledge from teachers is formulated as two parts: feature maps from the intermediate layers and final predictions.

To be specific, to transfer the knowledge from intermediate layers, we adopt the PKT [22] to match the probability distribution of the data in the feature space between teacher and student. Conventional feature distillation approaches (e.g., Fitnet [9] and attention maps [21]) intend to directly match the actual feature representations of teacher and student for a specific sample. Unlike the above methods, PKT transfers the probability distribution of samples in the feature space from the teacher to the student. Specifically, the conditional probability distribution models the pairwise affinity between two data samples in the feature space. Via minimizing the divergence between teacher's and student's probability distribution, the student is capable of creating similar geometry of feature space as the teacher.

For multiple teachers scenario, we first average the feature maps over all selected teachers and then calculate the PKT loss

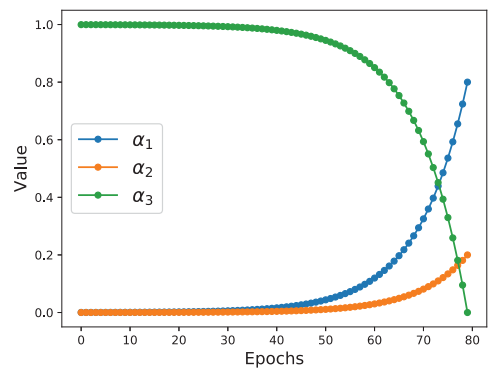

Fig. 3. Values of α_1, α_2, and α_3 at different training epochs.

by measuring the Kullback–Leibler (KL) divergence between two distributions as the following equation:

$$\mathcal{L}_{\text{PKT}} = D_{\text{KL}}(P \| Q) = \sum_{i=1}^{n_b} \sum_{j=1, j \neq i}^{n_b} p_{j|i} \log\Big(\frac{p_{j|i}}{q_{j|i}}\Big) \tag{7}$$

where $p_{j|i}$ and $q_{j|i}$ are the conditional probability distributions of the teacher and student with respect to sample x_i and x_j in the mini-batch. Equation (8) presents the calculation of $p_{j|i}$. Here, $\widetilde{f}_{x_i}$ and $\widetilde{f}_{x_j}$ are the averages of flattened feature maps over all selected teachers with sample x_i and x_j as inputs. $\cos(\widetilde{f}_{x_i}, \widetilde{f}_{x_j})$ measures the biased cosine similarity between $\widetilde{f}_{x_i}$ and $\widetilde{f}_{x_j}$ as shown in (9). Comparing with the general calculation of cosine similarity, the biased cosine similarity shifts the value to make sure that the conditional probability distribution is positive

$$p_{j|i} = \frac{\cos(\widetilde{f}_{x_i}, \widetilde{f}_{x_j})}{\sum_{k=1, k \neq j}^{n_b} \cos(\widetilde{f}_{x_k}, \widetilde{f}_{x_j})}, \tag{8}$$

$$\cos(\widetilde{f}_{x_i}, \widetilde{f}_{x_j}) = \frac{1}{2}\left[\frac{(\widetilde{f}_{x_i})^T (\widetilde{f}_{x_j})}{||\widetilde{f}_{x_i}||_2 ||\widetilde{f}_{x_j}||_2} + 1\right]. \tag{9}$$

By minimizing the $\mathcal{L}_{\text{PKT}}$, we encourage the student to learn feature representation with similar probability distribution as the teachers.

Meanwhile, we also utilize teachers' predictions as the soft labels and define the soft loss $\mathcal{L}_{\text{soft}} = 1/n_b \sum_1^{n_b} (\widetilde{y}_T - \hat{y}_S)^2$ as the distance between the averaged predictions $\widetilde{y}_T$ from all selected teachers and the prediction of student $\hat{y}_S$. In addition to the above two distillation losses, a general loss between student's prediction $\hat{y}_S$ and ground truth y is also adopted in student's training process, denoted as $\mathcal{L}_{\text{hard}} = 1/n_b \sum_1^{n_b} (\hat{y}_S - y)^2$. The final loss is defined as the combination of above three losses as follows:

$$\mathcal{L} = \alpha_1 * \mathcal{L}_{\text{hard}} + \alpha_2 * \mathcal{L}_{\text{soft}} + \alpha_3 * \mathcal{L}_{\text{PKT}} \tag{10}$$

where α_1, α_2, and α_3 are hyperparameters to balance the three losses. Fig. 3 illustrates the changing curve of α_1, α_2, and α_3 over different training epochs. Particularly, the weight α_3 for PKT loss is gradually decreased while the weights α_1 and α_2 for hard loss and soft loss are increased during the whole training process. The summation of α_1, α_2, and α_3 always equals to 1 at any training step. The motivation is that: at the early training

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

TABLE I
SUMMARY OF C-MAPSS DATASET

Dataset	FD001	FD002	FD003	FD004
Operating Conditions	1	6	1	6
Fault Modes	1	1	2	2
Train Traj.	100	260	100	249
Test Traj.	100	259	100	248

stage of student, the feature maps of the selected teachers could provide more information than a scalar, like soft label and ground truth. It would help the student to learn similar feature representations as the teachers first. As the training goes on, the weights of soft loss and hard loss would gradually increase, which allows the student to focus more on the regression target.

IV. EXPERIMENTS

In this section, we evaluate the effectiveness of our proposed approach on two real-world time series regression tasks, i.e., machine RUL prediction and indoor localization, which consists of six public datasets.

A. Experimental Setup

1) Machine RUL Prediction: We adopt the NASA's turbofan engine data for the RUL prediction task, which is also known as C-MAPSS dataset [33]. The objective is to accurately estimate the RUL of turbofan engines via 21 time series sensor measurements. It consists of four datasets, namely, FD001, FD002, FD003, and FD004. The operating conditions and fault modes of engines in each dataset are different, as shown in Table I. Each dataset contains several training and test trajectories. Each trajectory in training set represents a turbofan engine with different initial states and contains 21 time-series sensor measurements recording engine degradation process. On the contrary, the trajectories in test sets are the engine measurements at certain degradation stage. The objective of this dataset is to accurately estimate the RUL of turbofan engines via these sensor measurements. In our experiments, the training sets are further divided into training and validation set with a ratio of $9:1$ in terms of trajectories. For instance, we randomly select 90 trajectories from FD001 for model training and the rest ten trajectories for validation. Then, the same data preprocessing method has been applied to training, validation, and test sets as [20]. Moreover, same as previous works [19], [32], [34], the RMSE and Score function are employed as the evaluation metrics, which are defined as

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}, \tag{11}$$

$$\text{Score} = \begin{cases} \sum_{i=1}^{N}(e^{-\frac{\hat{y}_i - y_i}{13}} - 1), & \text{if } \hat{y}_i < y_i \\ \sum_{i=1}^{N}(e^{\frac{\hat{y}_i - y_i}{10}} - 1), & \text{otherwise} \end{cases} \tag{12}$$

where $\hat{y}_i$ and y_i are the predicted RUL and the ground truth RUL and N is the total number of samples. The lower RMSE and Score are, the better performance the model would achieve. Particularly, the Score function places more penalty on late prediction than early prediction, as the late prediction may cause worse catastrophe in real world applications.

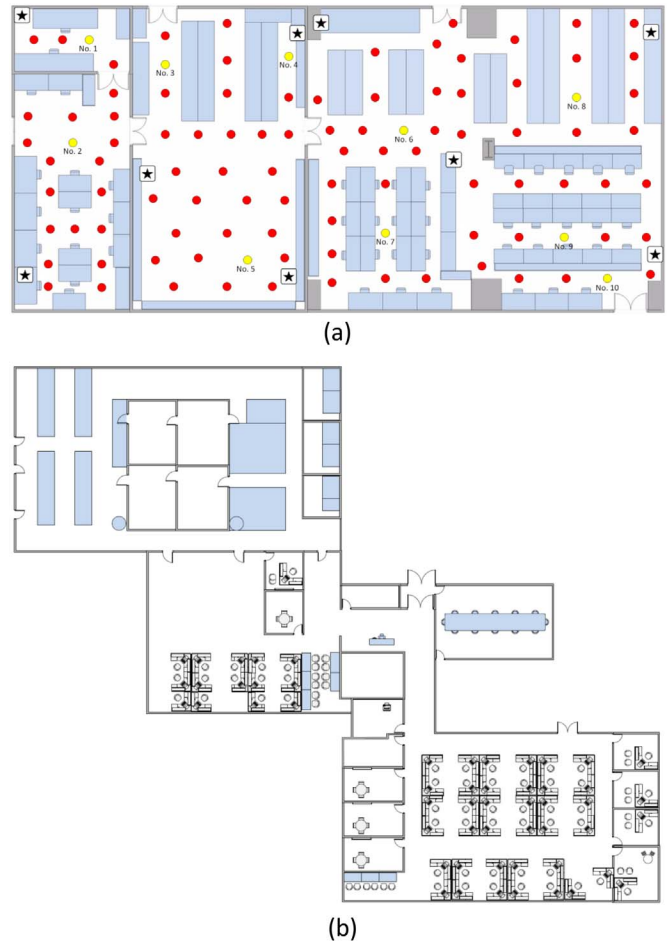

Fig. 4. Layout of two indoor localization environments: (a) lab and (b) office.

2) Indoor Localization: Another typical time series regression task that we utilized to evaluate our method is indoor localization with WiFi received signal strength indicator (RSSI) values [35]. It consists of two datasets collected under different settings. Specifically, the dataset was collected by using off-the-shelf WiFi routers in two different environments: a research lab (35.3×16.0 m) with nine WiFi routers and an office (55.0×50 m) with 20 WiFi routers as depicted in Fig. 4. To be specific, 102 reference points are selected in the research lab, which are uniformly distributed. At each point, 2000 consecutive RSSI values are collected from all nine routers [star symbols in Fig. 4(a)]. Then, data from 92 randomly selected reference points (red circles) are used for model training and the rest ten reference points (yellow circles) for testing. In the office, 353 uniformly distributed reference points are selected and 30 of them are randomly selected for testing and the rest for training. More detailed descriptions of the dataset can be found in [35]. We adopt mean localization error (MLE) as the

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

TABLE II
NETWORK DETAILS OF DDQN

Layers	Operations		Output Shape
Input	-		(None, 10, 64, 64)
Layer 1	Conv2D(32,7,3) + Relu		(None, 32, 20, 20)
Layer 2	Conv2D(64,5,2) + Relu		(None, 64, 8, 8)
Layer 3	Conv2D(64,3,1) + Relu		(None, 64, 6, 6)
Layer 4	Flatten		(None, 2304)
Layer 5	FC(2034, 512) + Relu	FC(2034,512) + Relu	(None, 512), (None, 512)
Layer 6	FC(512, 1)	FC(512, 10)	(None, 1), (None, 10)
Output	Aggregation as Equation (4)		(None, 10)

evaluation metric following [35]. The MLE is defined as

$$\text{MLE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}[(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2]} \tag{13}$$

where $(\hat{x}_i, \hat{y}_i)$ are predicted location coordinates and (x_i, y_i) are the ground truth.

3) Model Implementation: For fair comparison with existing works, we adopt the same network architectures for LSTM-based teacher and CNN-based student as [19], [20] for RUL prediction task. Particularly, the teacher model consists of five LSTM layers (32 hidden units in each layer) as the feature extractor and two fully connected layers as the regressor. The student model consists of three parallel dilated CNN branches as the feature extractor and also two fully connected layers (but with less hidden units) as the regressor. For the indoor localization task, we employ a model consisting of two-layer LSTM and two fully connected layers as the teacher (same as [35]) and use the same CNN-based student as RUL prediction task but only modify the last layer from single scalar output to a tuple output [(x, y) coordinates for the localization]. Moreover, we follow [36], [37], [38] and employ a three-layer CNN as the common encoder mentioned in Section III and two fully connected layers for each stream. Table II gives the details of the DDQN architecture. The output shape is calculated based on $M = 10$ and $n_b = 64$. "Conv2D(32,7,3)" refers to a 2-D convolutional operation with output_channels = 32, kennel_size = 7, and stride = 3. "FC(2034, 512)" refers to a fully connected layer with input = 2034 and output = 512. Layers 1 to 4 are the common CNN encoder. The left branch of layers 5 and 6 is the state-value stream and the right branch is the advantage stream. The final output is the aggregation of the two streams. Moreover, we conducted further analysis for different structures of DDQN encoder in Section IV-D.

4) Training Details: To generate the negatively correlated teachers pool, we first initialize M = 20 teachers with different random seeds and then simultaneously train them with the NCL strategy. Specifically, following [18] we configure the correlation penalty weight $\lambda = 0.5$ and set $n_b = 64$, learning rate $\text{lr} = 1\text{e}^{-3}$. The Adam optimizer is adopted during training. In RL-based KD stage, totally K teachers are selected out for each mini-batch. Note that student's performance varies with the number of K selected teachers for each set. To be specific, we set $K = 3$ for FD001 and FD003 in machine RUL prediction task and indoor localization task, $K = 5$ for FD002 and FD004. More experimental results regarding the K will be presented in Section IV-D. The student is trained with $\text{lr} = 1\text{e}^{-3}$ with a step decay schedule and Adam optimizer. For the DDQN optimization, we follow [31] and set $\gamma = 0.9$ and $\delta = 0.999$. The DDQN is trained with a batch size of 32, $\text{lr} = 1\text{e}^{-4}$ and Adam optimizer. All the experiments are conducted on a NVIDIA 2080Ti GPU.

B. Results and Discussion

1) Comparison With Benchmark Approaches: To verify the effectiveness of the proposed approach, extensive experiments have been conducted to compare our method with SOTA methods. Table III presents the evaluation results of different KD methods on two tasks. In particular, we compare student's performance under four scenarios: no teacher, single teacher, multiple teachers, and reinforced multiteacher. For no teacher case, we train the student with the ground truth only, denoted as "**Student Only**." We take it as the baseline to other KD methods. For the single teacher case, we employ the teacher with best performance on the validation set from our negatively correlated teachers pool to perform distillation with different methods. We compare three different feature distillation approaches including **AT** [21], **RKD-DA** [23], and **PKT** [22]. We also include the results from two recent SOTA works for C-MAPSS dataset targeting at transferring knowledge between disparate network architecture: **KDnet-RUL** [19] and **CA-KD** [20]. Meanwhile, we apply these two SOTA approaches in indoor localization task. Particularly, for **KDnet-RUL**, we follow [19] and apply sequential distillation upon identical architecture knowledge transferring. For **CA-KD**, we set the number of negative samples in contrastive learning to 512 same as [20]. Besides, we compare three widely used ensemble methods for multiteacher scenario: randomly selecting one teacher from all teachers (denoted as **Random-T** [25]), using fixed weights for all teachers (denoted as **W-Averaging** [17]), and using uniform distributed weights for all teachers (denoted as **U-Averaging** [7]). For **W-Averaging**, we assign weights based on their performance on validation set. The better performance a teacher can achieve, the higher weight it will be assigned to.

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

TABLE III
PERFORMANCE COMPARISON WITH BENCHMARK APPROACHES

Tasks		Machine RUL Prediction [33]								Indoor Localization [35]	
Scenario	Methods	FD001		FD002		FD003		FD004		Lab	Office
		RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score	MLE	MLE
No Teacher	Student Only	14.64	392.23	16.15	1281.07	15.34	602.15	17.38	1760.41	1.78	2.12
Single Teacher	Standard KD [7]	13.82	320.4	15.59	1131.33	14.16	521.16	16.86	1549.98	1.76	1.95
	AT [21]	13.91	344.05	15.38	1034.96	14.75	593.28	17.02	1570.05	1.70	2.01
	RKD-DA [23]	13.67	309.69	15.18	1008.52	14.04	480.25	16.55	1418.40	1.73	2.02
	PKT [22]	13.57	332.28	14.41	996.04	13.17	350.86	15.94	1291.87	1.66	1.92
	KDnet-RUL [19]	13.68	362.08	14.47	929.20	12.95	327.27	15.96	1303.19	1.51	1.83
	CA-KD [20]	13.41	293.82	14.23	975.96	12.95	325.29	15.85	1256.82	1.53	1.79
Multiple Teachers	Random-T [25]	14.15	314.85	15.41	990.02	14.09	450.72	16.89	1590.01	1.58	1.85
	W-Averaging [17]	13.76	309.22	15.56	1012.82	13.94	468.64	16.60	1460.99	1.60	1.89
	U-Averaging [7]	13.60	306.59	15.27	983.48	14.12	477.57	16.42	1308.48	1.58	1.87
Reinforced Multi-Teacher	Independent [11]	13.55	310.53	15.10	1054.82	13.72	426.76	16.53	1340.85	1.50	1.78
	Sequential [12]	13.71	312.81	15.03	972.74	13.87	404.48	16.41	1385.85	1.52	1.84
	Proposed	**13.07**	**288.82**	**14.22**	**901.74**	**12.82**	**311.55**	**15.71**	**1241.31**	**1.47**	**1.76**

The summation of all weights equals to 1. From Table III, some observations are as follows.

First, all KD methods, either using single teacher or multiple teachers, could enhance student's performance in both machine RUL prediction task and indoor localization task. It indicates the effectiveness of KD in model compression.

Second, in single teacher KD scenario, among three different feature distillation methods (i.e., AT, RKD-DA, and PKT), PKT outperforms other two feature distillation methods in terms of transferring knowledge from single teacher. Therefore, in our multiteacher scenario, PKT is employed to match the probability distribution of the feature maps between student and the average of the selected teachers. Besides, KDnet-RUL and CA-KD have superior performance than others for most of subsets in both tasks. One possible reason is that the knowledge transferring schemes in these two methods are specially designed for disparate network architectures. However, introducing the adversarial, sequential, or contrastive learning strategies on feature distillation would significantly increase training efforts.

Third, in multiple teachers KD scenario, the simple averaging strategy over all teachers (i.e., U-Averaging) could lead to better performance than the other two ensemble methods (Random-T and W-Averaging) for most of subsets in both machine RUL prediction task and indoor localization task. In other words, assigning more weights to a stronger teacher based on its performance on the whole training samples does not necessarily improve student's performance. This observation reveals the importance of properly choosing 'decent' teachers on instance level.

Lastly, the proposed RL-based KD method significantly and also consistently outperforms all aforementioned methods. The compact student trained with our approach can achieve average improvements of 5% and 15% in terms of RMSE and Score respectively compared to existing single-teacher KD methods, 7% and 16% in terms of RMSE and Score over existing multiteachers KD methods for machine RUL prediction tasks. Similarly, we observed average improvement of 9.6% in terms of MLE over existing single-teacher KD methods and 6.6% in terms of MLE over existing multiteachers KD methods for indoor localization task. These observations demonstrate the superiority of our method over other ensemble strategies and reveal that our method could help the student to better learn feature representations via properly selecting suitable teachers for different instances. Besides, our method eliminates the process of teacher selection based on prior experience, hence could be more efficient.

2) Different Teachers: Furthermore, we explicitly show how well three different types of teachers (as depicted in Fig. 2) could transfer the knowledge to a student. All these three types of teachers have exactly the same network architecture but with different training strategy. For the independent teachers, each individual was randomly initialized and trained independently. For the snapshot teachers, we adopted a cyclical learning rate schedule with warm restart method to improve the convergence rate and saved the intermediate network (snapshot) every ten epochs along single training process. Other training parameters for independent and sequential teachers, like learning rate, optimizer, and configuration of the RL-based KD stage, are the same as negatively correlated teachers for fair comparison. The last three rows in Table III show the student's performance with different types of teachers. On the one hand, comparing with other ensemble strategies the proposed RL-based teacher selection could also boost student's performance even with independent and sequential teachers. It reveals that our proposed RL-based teacher selection method can be applied to any other KD-related applications using multiple teachers. On the other hand, although our proposed NCL module requires more memory than sequential and independent strategies for simultaneously training multiple teachers, students trained with negatively correlated teachers have consistent superior performance on all datasets than other two types of teachers. This observation indicates the necessity of diversifying teachers for distilling more comprehensive knowledge about the data on multiteacher KD scenario.

C. Ablation Study

To investigate how each KD component contributes to student's final performance, we conducted the ablation study on machine RUL prediction task as shown in Fig. 5. Fig. 5(a)–5(d)

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ANALYSIS OF DIFFERENT STRUCTURES OF DDQN ENCODER

Layers of DDQN Encoder	Machine RUL Prediction								Indoor Localization	
	FD001		FD002		FD003		FD004		Lab	Office
	RMSE	Score	RMSE	Score	RMSE	Score	RMSE	Score	MLE	MLE
1 Layer	14.21	387.64	15.96	1164.54	14.87	586.31	17.02	1581.64	1.75	1.94
3 Layers **(Ours)**	13.07	288.82	14.22	901.74	12.82	311.55	15.71	1241.31	1.47	1.76
5 Layers	13.04	279.36	14.30	921.54	12.79	305.67	15.69	1245.61	1.48	1.76
7 Layers	13.06	292.44	14.19	894.20	12.84	325.14	15.65	1238.65	1.46	1.74

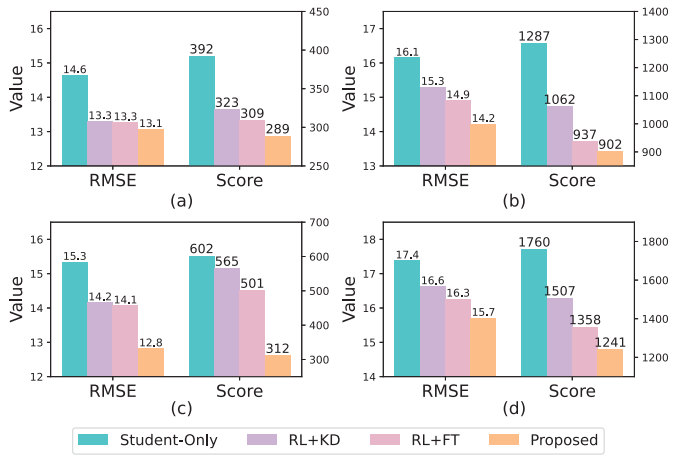

Fig. 5. Analysis of different components in the proposed method on C-MAPSS. (a) FD001; (b) FD002; (c) FD003; and (d) FD004.

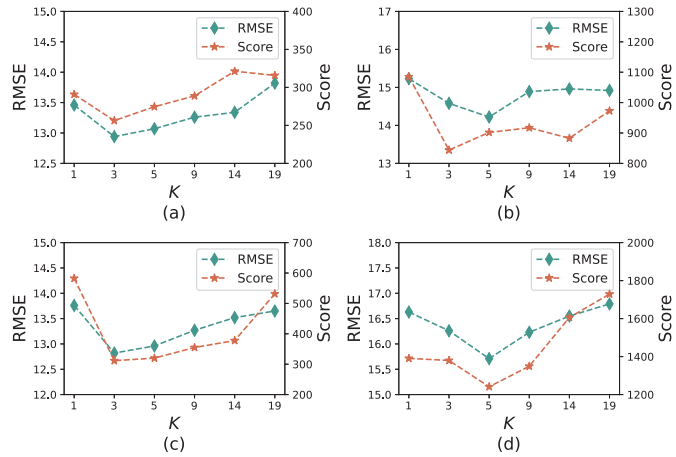

Fig. 6. Sensitivity of K selected teachers on C-MAPSS. (a) FD001; (b) FD002; (c) FD003; and (d) FD004.

shows the RMSE and Score results from FD001 to FD004. "RL+KD" represents the student trained with ground truth labels and soft labels for KD. Similarly, "RL+FT" means that the student is trained with ground truth labels and feature distillation with PKT. The teachers in "RL+KD" and "RL+FT" are selected via the proposed RL-based teacher selection method. It can be clearly seen from Fig. 5 that each component in KD could boost student's performance. It is also not surprising that "RL+FT" outperforms "RL+KD" since the soft label in the regression task is just a scalar, which in general is a probability distribution in classification tasks. The knowledge provided by a scalar is very limited. However, the feature distillation and soft labels from teacher are complementary to each other. Combining them together could yield a better performance as our proposed method.

D. Sensitivity Analysis

1) Number of Selected Teachers: One of the key parameters in our proposed reinforcement learning based KD method is the number of K teachers selected for each batch training. We conducted a detailed investigation on how the K value affects student's performance on machine RUL prediction task. The results for four subsets from C-MAPSS are shown in Fig. 6.

The optimal K for each subset is different. For simple datasets (e.g., FD001 and FD003), a set of $K = 3$ teachers would produce good student. However, for the complex datasets (e.g., FD002 and FD004), they need more teachers ($K = 5$) to participate in distillation stage. Furthermore, student's performance would gradually decrease when there are more teachers involving into distillation. The reason is that involving unknowledgable teachers for certain specific samples would introduce noise into student's learning process.

2) Analysis of Different Architectures of DDQN Encoder: As DDQN network plays a critical role in RL-based teacher selections process, we further analysis how different architectures of DDQN encoder affect final performance. Table IV summaries student's performance with different encoders on the two datasets. We can see that shallow encoder (i.e., only using a single CNN layer as encoder) performs worst as expected as its compact architecture severely impacts the learning efficiency of teacher selection policy. On the other hand, for most of datasets, student's performance would slightly get improved when increasing model complexity of DDQN encoder. However, the improvement is very marginal, indicating that our proposed RL-based teacher selection strategy is not very sensitive to the design of DDQN.

V. CONCLUSION

In this article, we show that teachers trained via negative correlation learning are more comprehensive and suitable for multiteacher KD scenario. A reinforcement learning based teacher selection approach is then proposed to dynamically select different teachers to participate in the distillation for different

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

training samples. The experiment results demonstrate that our proposed method significantly outperforms other SOTA KD methods and prove the effectiveness of our method for time series regression.

In our future work, instead of explicitly predefining student's architecture, we intend to apply RL-based method to automatically generate suitable student networks under some application-specific constraints, such as real-time requirements in autonomous driving or utilization of hardware accelerators [39]. Besides, considering the additional computational resources required on single training device for our reinforced KD method, distributed on-device distillation [40] with multiple teachers could also be a potential extension.

References

[1] Q. Hua, M. H. Nielsen, Z. An, and M. Shen, "Small-scale dataset generalized NLOS identification for multi-scene indoor localization," *IEEE Trans. Artif. Intell.*, vol. 1, pp. 1–14, Mar. 2022.

[2] A. Tittaferrante and A. Yassine, "Multiadvisor reinforcement learning for multiagent multiobjective smart home energy control," *IEEE Trans. Artif. Intell.*, vol. 3, no. 4, pp. 581–594, Aug. 2022.

[3] B.-L. Lu, Z.-H. Liu, H.-L. Wei, L. Chen, H. Zhang, and X.-H. Li, "A deep adversarial learning prognostics model for remaining useful life prediction of rolling bearing," *IEEE Trans. Artif. Intell.*, vol. 2, no. 4, pp. 329–340, Aug. 2021.

[4] R. Zhao, R. Yan, Z. Chen, K. Mao, P. Wang, and R. X. Gao, "Deep learning and its applications to machine health monitoring," *Mech. Syst. Signal Process.*, vol. 115, pp. 213–237, Jan. 2019.

[5] T. Liang, J. Glossner, L. Wang, S. Shi, and X. Zhang, "Pruning and quantization for deep neural network acceleration: A survey," *Neurocomputing*, vol. 461, pp. 370–403, Oct. 2021.

[6] T. Elsken, J. H. Metzen, and F. Hutter, "Neural architecture search: A survey," *J. Mach. Learn. Res.*, vol. 20, no. 1, pp. 1997–2017, 2019.

[7] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," in *Proc. NIPS Deep Learn. Represent. Learn. Workshop*, vol. 1050, 2015, p. 9.

[8] Y. Kim, D. Cho, K. Han, P. Panda, and S. Hong, "Domain adaptation without source data," *IEEE Trans. Artif. Intell.*, vol. 2, no. 6, pp. 508–518, Dec. 2021.

[9] A. Romero, N. Ballas, S. E. Kahou, A. Chassang, C. Gatta, and Y. Bengio, "FitNets: Hints for thin deep nets," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.

[10] J. H. Cho and B. Hariharan, "On the efficacy of knowledge distillation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 4794–4802.

[11] A. J. C. Sharkey, "On combining artificial neural nets," *Connect. Sci.*, vol. 8, nos. 3–4, pp. 299–314, 1996.

[12] G. Huang, Y. Li, G. Pleiss, Z. Liu, J. E. Hopcroft, and K. Q. Weinberger, "Snapshot ensembles: Train 1, get M for free," 2017, *arXiv:1704.00109*.

[13] Y. Liu, K. Wang, G. Li, and L. Lin, "Semantics-aware adaptive knowledge distillation for sensor-to-vision action recognition," *IEEE Trans. Image Process.*, vol. 30, pp. 5573–5588, 2021.

[14] Y. Tian, D. Krishnan, and P. Isola, "Contrastive representation distillation," 2019, *arXiv:1910.10699*.

[15] M.-C. Wu, C.-T. Chiu, and K.-H. Wu, "Multi-teacher knowledge distillation for compressed video action recognition on deep neural networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Piscataway, NJ, USA: IEEE Press, 2019, pp. 2202–2206.

[16] W.-C. Kao, H.-X. Xie, C.-Y. Lin, and W.-H. Cheng, "Specific expert learning: Enriching ensemble diversity via knowledge distillation," *IEEE Trans. Cybern.*, vol. 53, no. 4, pp. 2494–2505, Apr. 2023.

[17] Y. Chebotar and A. Waters, "Distilling knowledge from ensembles of neural networks for speech recognition," in *Proc. Interspeech*, 2016, pp. 3439–3443.

[18] Y. Liu and X. Yao, "Simultaneous training of negatively correlated neural networks in an ensemble," *IEEE Trans. Syst., Man, Cybern., B (Cybern.)*, vol. 29, no. 6, pp. 716–725, Dec. 1999.

[19] Q. Xu, Z. Chen, K. Wu, C. Wang, M. Wu, and X. Li, "KDnet-RUL: A knowledge distillation framework to compress deep neural networks for machine remaining useful life prediction," *IEEE Trans. Ind. Electron.*, vol. 69, no. 2, pp. 2022–2032, Feb. 2022.

[20] Q. Xu, Z. Chen, M. Ragab, C. Wang, M. Wu, and X. Li, "Contrastive adversarial knowledge distillation for deep model compression in time-series regression tasks," *Neurocomputing*, vol. 485, pp. 242–251, May 2021.

[21] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," in *Proc. 5th Int. Conf. Learn. Representations (ICLR)*, 2017.

[22] N. Passalis and A. Tefas, "Learning deep representations with probabilistic knowledge transfer," in *Proc. Eur. Conf. Comput. Vision (ECCV)*, 2018, pp. 268–284.

[23] W. Park, D. Kim, Y. Lu, and M. Cho, "Relational knowledge distillation," in *Proc. IEEE Conf. Comput. Vision Pattern Recognit.*, 2019, pp. 3967–3976.

[24] S. Park and N. Kwak, "Feature-level ensemble knowledge distillation for aggregating knowledge from multiple networks," in *Proc. Eur. Conf. Artif. Intell. (ECAI)*, Amsterdam, The Netherlands: IOS Press, 2020, pp. 1411–1418.

[25] T. Fukuda, M. Suzuki, G. Kurata, S. Thomas, J. Cui, and B. Ramabhadran, "Efficient knowledge distillation from an ensemble of teachers," in *Proc. Interspeech*, 2017, pp. 3697–3701.

[26] S. Fan, X. Zhang, and Z. Song, "Reinforced knowledge distillation: Multi-class imbalanced classifier based on policy gradient reinforcement learning," *Neurocomputing*, vol. 463, pp. 422–436, Nov. 2021.

[27] J. Wang et al., "Multi-constraint molecular generation based on conditional transformer, knowledge distillation and reinforcement learning," *Nature Mach. Intell.*, vol. 3, no. 10, pp. 914–922, 2021.

[28] A. Tsantekidis, N. Passalis, and A. Tefas, "Diversity-driven knowledge distillation for financial trading using deep reinforcement learning," *Neural Netw.*, vol. 140, pp. 193–202, Aug. 2021.

[29] S. Liang et al., "Reinforced iterative knowledge distillation for cross-lingual named entity recognition," in *Proc. 27th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, 2021, pp. 3231–3239.

[30] F. Yuan et al., "Reinforced multi-teacher selection for knowledge distillation," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 14284–14291.

[31] K. Wu, M. Wu, J. Yang, Z. Chen, Z. Li, and X. Li, "Deep reinforcement learning boosted partial domain adaptation," in *Proc. 30th Int. Joint Conf. Artif. Intell. (IJCAI)*, Z.-H. Zhou, Ed., Aug. 2021, pp. 3192–3199, doi: 10.24963/ijcai.2021/439.

[32] Z. Chen, M. Wu, R. Zhao, F. Guretno, R. Yan, and X. Li, "Machine remaining useful life prediction via an attention based deep learning approach," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2521–2531, Mar. 2021.

[33] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in *Proc. Int. Conf. Prognostics Health Manage.*, Piscataway, NJ, USA: IEEE Press, 2008, pp. 1–9.

[34] G. S. Babu, P. Zhao, and X.-L. Li, "Deep convolutional neural network based regression approach for estimation of remaining useful life," in *Proc. Int. Conf. Database Syst. Adv. Appl.*, 2016, pp. 214–228.

[35] Z. Chen, H. Zou, J. Yang, H. Jiang, and L. Xie, "WiFi fingerprinting indoor localization using local feature-based deep LSTM," *IEEE Syst. J.*, vol. 14, no. 2, pp. 3001–3010, Jun. 2020.

[36] Z. Wang, T. Schaul, M. Hessel, H. Hasselt, M. Lanctot, and N. Freitas, "Dueling network architectures for deep reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2016, pp. 1995–2003.

[37] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[38] H. Van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 30, no. 1, pp. 2094–2100, 2016.

[39] S. Shaw, K. Tyagi, and S. Zhang, "Teacher–student knowledge distillation for radar perception on embedded accelerators," 2023, *arXiv:2303.07586*.

[40] I. Bistritz, A. Mann, and N. Bambos, "Distributed distillation for on-device learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 22593–22604.

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.

Qing Xu received the B.Eng. degree in measuring control technology and instruments and the M.Eng. degree in instrument science and technology from Southeast University, Nanjing, China, in 2010 and 2015, respectively. He is currently working toward the Ph.D. degree with Nanyang Technological University (NTU), Singapore.

He is a Senior Research Engineer with the Institute for Infocomm Research, Agency for Science, Technology and Research (A*STAR), Singapore. His research interests include deep learning, transfer learning, model compression, and related applications.

Keyu Wu received the B.Eng. degree in bioengineering from the National University of Singapore, Singapore, in 2013, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2020.

She is currently a Scientist with the Institute for Infocomm Research, Agency for Science, Technology and Research (A*STAR), Singapore. Her research interests include reinforcement learning, transfer learning, unsupervised learning, path planning, and autonomous navigation.

Min Wu (Senior Member, IEEE) received the B.S. degree in computer science from the University of Science and Technology of China (USTC), Hefei, China, in 2006, and the Ph.D. degree in computer science from Nanyang Technological University (NTU), Singapore, in 2011.

He is currently a Senior Scientist with the Data Analytics Department, Institute for Infocomm Research, Agency for Science, Technology and Research (A*STAR), Singapore. He received the Best article Awards in InCoB 2016 and DASFAA 2015. He also won the IJCAI competition on repeated buyers prediction in 2015. His current research interests include machine learning, data mining, and bioinformatics.

Kezhi Mao (Member, IEEE) received the B.Eng., M.Eng., and Ph.D. degrees from the University of Jinan, Jinan, China, Northeastern University, Shenyang, China, and the University of Sheffield, Sheffield, England, in 1989, 1992, and 1998, respectively.

Since then, he has been working with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, and is now an Associate Professor. His research covers a couple of subfields of artificial intelligence (AI), including machine learning, image processing, natural language processing, and information fusion. Over the years, he has developed novel algorithms and frameworks to address various problems in AI. As a strong advocate of translational research, he has collaborated with government agencies and hospitals and developed a couple of prototypes of AI systems for image processing and natural language processing.

Dr. Mao now serves as Member of Editorial Board of neural networks, an Academic Editor of computational intelligence and neuroscience, and the General Chair and a General Co-Chair of a number of interna tional conferences.

Xiaoli Li (Senior Member, IEEE) received the B.Eng. degree from the School of Computer Science, Shanxi University, Shanxi, China, and Ph.D. degree from the Institute of computing technology, Chinese Academy of Science, China.

He is currently a Principal Scientist with the Institute for Infocomm Research, A*STAR, Singapore. He also holds Adjunct Full Professor position with Nanyang Technological University, Singapore. He has published more than 270 high quality articles and won eight best article/benchmark competition awards. His research interests include AI, machine learning, data mining, and bioinformatics.

Dr. Li has been serving as an Area Chair/Senior PC member in leading AI and data mining related conferences (including IJCAI, AAAI, KDD, and ICDM). He is currently serving as the Editor-in-Chief of World Scientific Annual Review of Artificial Intelligence, and an Associate Editor of IEEE TRANSACTIONS ON ARTIFICIAL INTELLIGENCE and *Machine Learning with Applications* (Elsevier).

Zhenghua Chen (Senior Member, IEEE) received the B.Eng. degree in mechatronics engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2011, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University (NTU), Singapore, in 2017.

Currently, he is a Scientist and Lab Head of the Institute for Infocomm Research, and an Early Career Investigator with Centre for Frontier AI Research (CFAR), Agency for Science, Technology and Research (A*STAR), Singapore. His research interests include data-efficient and model-efficient learning with related applications in smart city and smart manufacturing.

Dr. Chen has won several competitive awards, such as First Place Winner for CVPR 2021 UG2+ Challenge, A*STAR Career Development Award, First Runner-Up Award for Grand Challenge at IEEE VCIP 2020, and Best article Award at IEEE ICIEA 2022 and IEEE SmartCity 2022. He serves as an Associate Editor for IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT, IEEE TRANSACTIONS ON INDUSTRIAL CYBER-PHYSICAL SYSTEMS, IEEE SENSORS JOURNAL, *Springer Discover Artificial Intelligence*, and *Elsevier Neurocomputing*. He is currently the Chair of IEEE Sensors Council Singapore Chapter.

Authorized licensed use limited to: Indian Institute of Technology Patna. Downloaded on February 07,2026 at 12:34:12 UTC from IEEE Xplore. Restrictions apply.